



INSTITUTE OF TECHNOLOGY OF
CAMBODIA

AIIS
Department of Applied Mathematics and Statistics

Student Information Management



Subject: Data Structure and Programming 2

Lecturer: Ms. CHOM Sreylam

Group: 06

Academic Year: 2025 – 2026



Linked List Queue



Group Members

1

Hin Panha

2

Hor Soue Hak

3

Kun Virak

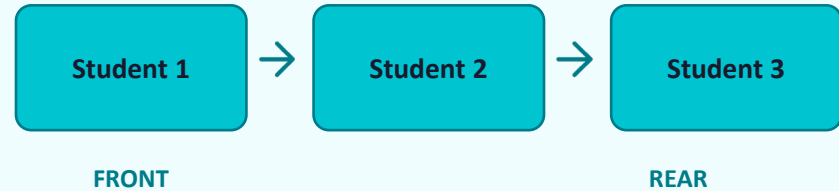


Data Structure

Student Node

id
name
age
gender
department
major
year
province
phone
birth
email
→ next

Queue (Linked List)



- Nodes stored in memory (heap)
- front & rear track the queue
- Each node links to the next
- CSV saves data permanently



Program Start



Open CSV File

Reads
Students_Information.csv



Load into Queue

Each row → a new Student
node



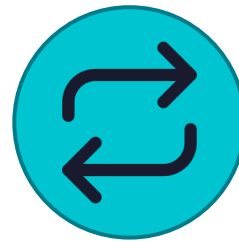
Show Menu

User picks an action (1–6)



Run Action

Execute the chosen function



Loop Again

Repeat until user exits (6)



Loop continues until user selects Exit (6)

Menu Options

1



Add Student

4



Search

2



Remove Student

5



Update Info

3



Display All

6



Exit

+ Add Student



1. Enter Student ID



2. Check if ID already exists



3. Duplicate ID? → Stop & warn



4. Enter all student info



5. Add node to end of queue



6. Save to CSV file



Always saved to CSV after adding

Fields Collected

ID • Name • Age

Gender • Department

Major • Year

Province • Phone

Birth • Email



Remove



Search



Remove Student



Enter Student ID to delete



Search through the queue



Adjust node pointers



Free memory (delete node)



Update REAR if needed



Save to CSV



Search Student



Enter Student ID to find



Walk through queue nodes



Compare each node's ID



Found → show all details



Not found → print error



Display & Sort



Bubble Sort (A → Z)

1. Compare name with next node
2. If out of order → swap ALL fields
3. Repeat until no more swaps
4. List is now sorted A → Z
5. Then display the table



Table Output

ID	Name	Age	Gender	Dept
----	------	-----	--------	------

Major	Year	Province	Phone	Birth	Email
-------	------	----------	-------	-------	-------

Setw() aligns each column evenly



Sorted alphabetically first



No file save needed for display



Update Student



Enter ID to find student



Search queue with findStudent()



Not found? → show error & stop



Show update menu (options 1–11)



Pick which field to change



Change saved to CSV file

11 Updatable Fields

ID

Year

Name

Province

Age

Phone

Gender

Birth

Department

Email

Major



File Storage (CSV)



SaveFile()

- Open CSV file for writing
- Write header row first
- Loop all nodes in queue
- Write each field with commas
- Close the file



DataIntoQueue()

- Open CSV file for reading
- Skip header row (getline)
- Read comma-separated fields
- Create new Student node
- Add node to the queue

CSV Example

ID,Name,Age,Gender,Department,Major,Year,Province,Phone,Birth,Email

e20241363,Hin Panha,19,Male,AMS,Data Science,2,Battambang,088 9157 750,06/12/2006,panhaahinn@example.com

Conclusion



Linked List

Stores students as connected nodes



Search

Find any student by ID quickly



CSV File

Saves & loads data from disk



Sort

Display sorted A→Z by name



CRUD

Add, Remove, Search, Update



Loop

Menu runs until user exits (6)

The background is a photograph of a modern, multi-story building with large windows and balconies. A teal-colored overlay covers the entire image. In the center, there is a large teal circle with a dark blue center. Inside the dark blue center, the words "Thank You" are written in white. Scattered around the teal circle are several small teal dots and heart-shaped icons. The building in the background has a sign that reads "INSTITU" and some Thai text. A yellow truck and a white car are parked in front of the building, and a few people are walking on the sidewalk.

Thank
You